Lab 06: Arrays

CSE 4108

Structured Programming I Lab

Lab Tasks

1. Repeat After Me:

Write a C program that takes an integer as an input and shows which digits (if any) were repeated.

Sample input:

```
Enter a number: 939577
```

Sample output:

```
Repeated digit(s): 7 9
```

2. Repeating Again:

Modify the **"Repeat After Me"** program so that the user can enter more than one number to be tested for repeated digits, considering all numbers combined. But this time, it will print the number of times each digit has repeated instead of only the repeating digits. The program should terminate when the user enters a number that's equal to 0.

Sample input:

```
Enter a number: 939577 5424 12345 0
```

Sample output:

```
digits         :    0 1 2 3 4 5 6 7 8 9
Occurrences    :    0 1 2 2 3 3 0 2 0 2
```

3. Light Spiral:

After staying home for two years, Light Origami lost his abilities to function as a normal human. He now reads all matrices in a clockwise spiral way. His friends are trying to understand how Light looks at the world.

Help his friends by writing a program in C that would scan an integer n from the user and then scan an n×n matrix. The output will be the matrix printed in Spiral form.

You may use Variable Length Arrays (VLAs) or consider that n will not be more than 10.

Sample input:

```
Enter dimension N: 4
Enter Matrix [4x4]:
1 2 3 4
12 13 14 5
11 16 15 6
10 9 8 7
```

Sample output:

```
Matrix that Light reads[4x4]:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

4. Neo from Matrix:

In the world of Matrix, Neo possesses super human abilities of dodging fast bullets by slowing down time. However, he is really bad with matrix multiplication.

Help Neo by writing a program in C that would scan an integer N from the user and then scan two N×N matrices. The it will show the product of the two matrices.

You may use VLAs or consider that N will not be more than 10.

Sample input:

```
Enter dimension N: 3
Enter Matrix A [3x3]:
1.1 2.2 3.3
4.4 5.5 6.6
7.7 8.8 9.9
Enter Matrix B [3x3]:
1 2 3
4 5 6
7 8 9
```

Sample output:

```
Resultant Matrix Product [3x3]:
33 39.6 46.2
72.6 89.1 105.6
```

```
112.2 138.6 165
```

5. Palindrome:

A palindrome is a word which remains the same while reading forward or backward. In other words, it is mirrored in the middle. For instance, the word “mom” is a palindrome.

Sample input:

```
Enter a word: abcba
```

Sample output:

```
palindrome
```

Sample input:

```
Enter a word: abcbc
```

Sample output:

```
Not palindrome
```